```
import java.util.Arrays;
import java.util.*;
public class A_SingleDimensionalArray {
        public static void main(String args[]) {
            int intArray[] = {10, 9, 8, 7};

            System.out.println("\nint Array..."+ intArray);

            //0,1,2,3 i = 4  -- 4<4 false
            for (int i = 0; i < intArray.length; i++) {
               System.out.print(intArray[i] + " ");
            }

            System.out.println("\n printing Array name using reference variable "+intArray );
            System.out.println("\n using toString" + Arrays.toString(intArray));

            System.out.println("\n printing array size  " + intArray.length);

          }

}




public class B_SingleDimAssigningArray {

   public static void main(String[] args) {
      int[] i = new int[3];
      int[] j;

      i[0] = 1;
      i[1] = 2;
      i[2] = 3;


      j = i;

      System.out.println("\n");
      for (int no = 0; no < 3; no++) {
         System.out.println(j[no] + " ");
      }

      //before chnaging the value of j array 0th index = 1
      j[0] =  100;
```

```
        System.out.println("\n");
        for (int no = 0; no < 3; no++) {
           System.out.println(j[no] + " ");
        }



        System.out.println("\n");
        for (int no = 0; no < 3; no++) {
           System.out.println(i[no] + " ");
        }

    }
}
```

```
import java.util.Arrays;

public class C_SingleDimArraysCopy{

    public static void main(String[] args) {

        System.out.println("\nArray a:\n");
        int[] a = {5, 4, 7, 3, 6};
        printArray(a);

        System.out.println("\n\nPrinting array variable " + a);

        int[] b = a; // b = 111

        System.out.println("\n\nArray b:\n");
        printArray(b);//

        b[0] = 1;
        b[1] = 2;

        System.out.println("\n\nAfter making changes in the first two elements of b:\n");

        System.out.println("\nArray b:\n");
        printArray(b);//

        System.out.println("\n\nArray a also changes:\n");
        printArray(a);
```

```
        // Creating a copy using Arrays.copyOf()
        int[] c = Arrays.copyOf(a, 5);


        System.out.println("\n\nAfter copying Array a to c, Array c:\n");
        printArray(c);

        System.out.println("\n\nAfter making changes in the first two elements of c:\n");

        c[0] = 10;
        c[1] = 20;

        System.out.println("\nArray c:\n");
        printArray(c);

        System.out.println("\n\nArray a has no changes:\n");
        printArray(a);

        System.out.println();
    }

    public static void printArray(int[] a) {
        for (int i = 0; i < a.length; i++) {
            System.out.print(a[i] + " ");
        }
    }
}
```

```
import java.util.Arrays;

public class C_SingleDimArraysCopy {

    public static void main(String[] args) {

        System.out.println("\nArray a:\n");
        int[] a = {5, 4, 7, 3, 6};
        printArray(a);

        System.out.println("\n\nPrinting array variable " + a);

        int[] b = a; // b = 111
```

```
        System.out.println("\n\nArray b:\n");
        printArray(b);//

        b[0] = 1;
        b[1] = 2;

        System.out.println("\n\nAfter making changes in the first two elements of b:\n");

        System.out.println("\nArray b:\n");
        printArray(b);//

        System.out.println("\n\nArray a also changes:\n");
        printArray(a);

        // Creating a copy using Arrays.copyOf()
        int[] c = Arrays.copyOf(a, 5);


        System.out.println("\n\nAfter copying Array a to c, Array c:\n");
        printArray(c);

        System.out.println("\n\nAfter making changes in the first two elements of c:\n");

        c[0] = 10;
        c[1] = 20;

        System.out.println("\nArray c:\n");
        printArray(c);

        System.out.println("\n\nArray a has no changes:\n");
        printArray(a);

        System.out.println();
    }

    public static void printArray(int[] a) {
        for (int i = 0; i < a.length; i++) {
            System.out.print(a[i] + " ");
        }
    }
}



import java.util.Arrays;

public class D_SingleDimRefType {
    public static void main(String[] args) {
```

```
        // Creating StringBuilder objects
        StringBuilder sb1 = new StringBuilder("Java");
        StringBuilder sb2 = new StringBuilder("Python");
        StringBuilder sb3 = new StringBuilder("C");

        //array of string builder objects
        StringBuilder[] stringBuilderArray = { sb1, sb2, sb3 };

        System.out.println(Arrays.toString(stringBuilderArray));
    }
}
```

```
import java.util.Arrays;

public class E_SingleDimRefTypeShallowCopy {

            public static void main(String[] args) {
                // Creating StringBuilder objects
        StringBuilder sb1 = new StringBuilder("Java");
        StringBuilder sb2 = new StringBuilder("Python");
        StringBuilder sb3 = new StringBuilder("C");

        // Creating the original array
        StringBuilder[] original = { sb1, sb2, sb3 };

        // Shallow Copy
        StringBuilder[] copy = Arrays.copyOf(original, original.length);

        // Modifying the object through the copied array
        copy[0].append(" Programming");

        // Printing both arrays
        System.out.println("Original Array : " + Arrays.toString(original));
        System.out.println("Copied Array   : " + Arrays.toString(copy));

            }

}
```

```
import java.util.Arrays;

public class F_SingleDimRefTypeDeepCopy {
```

```
    public static void main(String[] args) {

        // Creating StringBuilder objects
        StringBuilder sb1 = new StringBuilder("Java");
        StringBuilder sb2 = new StringBuilder("Python");
        StringBuilder sb3 = new StringBuilder("C");

        //System.out.println(sb3.toString());


        StringBuilder[] original = { sb1, sb2, sb3 };

       //1d - ref type - Shallow copy
        StringBuilder[] copy = Arrays.copyOf(original, original.length);

        //deep copy
        for (int i = 0; i < copy.length; i++) {
                //copy[0] = 1001
           copy[i] = new StringBuilder(copy[i].toString());
           //  copy[0] = new StringBuilder(copy[0].toString());

        }

        //copy[0].toString() = take first string builder object and
        //convert it into String object
        //and assigns the ref in copy[0]


        // Modifying the copied object
        copy[0].append(" Programming");

        // Printing both arrays
        System.out.println("Original Array : " + Arrays.toString(original));
        System.out.println("Copied Array   : " + Arrays.toString(copy));
    }
}
```

```
import java.util.Scanner;
import java.util.Arrays;

public class G_TwoDimArray {

    public static void main(String args[]) {
```

```
Scanner scanner = new Scanner(System.in);
int[][] tArray = new int[2][2]; // 2 rows and 2 columns

int original1[][] = {{1, 2, 3}, {2, 4, 5}, {4, 4, 5}};


//printing 2D array
for (int i = 0; i < 3; i++) {
   for (int j = 0; j < 3; j++) {
      System.out.print(original1[i][j] + " ");
   }
   System.out.println();
}

// Taking input from the user for tArray
System.out.println("Enter 4 elements for the 2x2 array");
for (int i = 0; i < 2; i++) {
   for (int j = 0; j < 2; j++) {
      System.out.print("Enter element at position [" + i + "][" + j + "]: ");
      tArray[i][j] = scanner.nextInt();
   }
}

System.out.println("\nThe 2x2 array is:");
for (int i = 0; i < 2; i++) {
   for (int j = 0; j < 2; j++) {
      System.out.print(tArray[i][j] + " ");
   }
   System.out.println(); // New line after each row
}

System.out.println("two dimensional array length");

System.out.println("Row count" + tArray.length);
System.out.println("columncount" + tArray[0].length);

for (int i = 0; i < tArray.length; i++) //row count
{
   for (int j = 0; j < tArray[i].length; j++) //column count in first row
   {
      System.out.print(tArray[i][j]);
      System.out.print("  ");
   }

   System.out.println();
}
```

```
        System.out.println("printing 2d array using variable " + Arrays.deepToString(tArray));

        System.out.println("Printing 2 d array using for each");
        for (int[] row : tArray) {
           for (int num : row) {
              System.out.print(num + " ");
           }
           System.out.println(); // New line after each row
        }

     }

}


import java.util.Arrays;

public class H_TwoDimShallowCopy {

            public static void main(String args[]) {
               int[][] original1 = {
                  {1, 2, 3},
                  {2,4,5},
                  {4,4,5}
               };


               // Copying the 2D array using Arrays.copyOf()
               int[][] copy = Arrays.copyOf(original1, original1.length);


               copy[0][0] = 100;
               // Printing  copied arrays using Arrays.deepToString()
               System.out.println("Copied Array: " + Arrays.deepToString(copy));
               System.out.println("Original Array: " + Arrays.deepToString(original1));


            }

}
```

```
import java.util.Arrays;
public class I_TwoDimArrayDeepCopy {

                public static void main(String[] args) {

                  int[][] original1 = {
                          {1, 2, 3},
                                {2, 4, 5},
                                {4, 4, 5}
                  };


                  int[][] copy = new int[original1.length][];

                  //Deep copy
                  for (int i = 0; i < original1.length; i++) {
                     copy[i] = Arrays.copyOf(original1[i], original1[i].length);
                  }

                  copy[0][0] = 100;

                  System.out.println("Copied Array   : " + Arrays.deepToString(copy));
                  System.out.println("Original Array : " + Arrays.deepToString(original1));
                }

        }



class J_TwoDimRefType
{
   public static void main(String[] args)
   {
      // Creating StringBuilder objects
      StringBuilder sb1 = new StringBuilder("Java");
      StringBuilder sb2 = new StringBuilder("Python");
      StringBuilder sb3 = new StringBuilder("C");
      StringBuilder sb4 = new StringBuilder("C++");
      StringBuilder sb5 = new StringBuilder("Spring");
      StringBuilder sb6 = new StringBuilder("Angular");

      // Creating the original 2D array
      StringBuilder[][] courses = {
                        { sb1, sb2, sb3 },
                        { sb4, sb5, sb6 }
                      };

      // Printing the elements
```

```
        System.out.println("2D Reference Type Array:");

        for (int i = 0; i < courses.length; i++)
        {
           for (int j = 0; j < courses[i].length; j++)
           {
              System.out.print(courses[i][j] + "\t");
           }
           System.out.println();
        }
     }
}


import java.util.Arrays;
public class K_TwoDimRefTypeShallowCopy {

              public static void main(String[] args)
              {
                 // Creating StringBuilder objects
                 StringBuilder sb1 = new StringBuilder("Java");
                 StringBuilder sb2 = new StringBuilder("Python");
                 StringBuilder sb3 = new StringBuilder("C");
                 StringBuilder sb4 = new StringBuilder("C++");
                 StringBuilder sb5 = new StringBuilder("Spring");
                 StringBuilder sb6 = new StringBuilder("Angular");

                 // Creating the original 2D array
                 StringBuilder[][] original = {
                                          { sb1, sb2, sb3 },
                                          { sb4, sb5, sb6 }
                                        };

                 // Creating a shallow copy
                 StringBuilder[][] shallow = Arrays.copyOf(original, original.length);

                 // Modifying an object through the shallow copy
                 shallow[0][0].append(" Full Stack----");

                 // Printing the Original Array
                 System.out.println("Original Array:");
                 printArray(original);

                 // Printing the Shallow Copy
                 System.out.println("\nShallow Copy:");
                 printArray(shallow);
              }
```

```
    // Method to print a 2D array
    static void printArray(StringBuilder[][] array)
    {
        for (int i = 0; i < array.length; i++)
        {
            for (int j = 0; j < array[i].length; j++)
            {
                System.out.print(array[i][j] + "\t");
            }
            System.out.println();
        }
    }
}
```

```
public class L_TwoDimDeepCopy {

    public static void main(String[] args)
    {
        // Creating StringBuilder objects
        StringBuilder sb1 = new StringBuilder("Java");
        StringBuilder sb2 = new StringBuilder("Python");

        StringBuilder sb3 = new StringBuilder("Spring");
        StringBuilder sb4 = new StringBuilder("Angular");

        // Creating the original 2D array
        StringBuilder[][] original = {
                                        { sb1, sb2 },
                                        { sb3, sb4 }
                                      };

        // Creating a deep copy
        StringBuilder[][] deep = new StringBuilder[original.length][];

        for (int i = 0; i < original.length; i++)
        {
            // Create a new inner array
            deep[i] = new StringBuilder[original[i].length];

            for (int j = 0; j < original[i].length; j++)
            {
                // Create a new StringBuilder object
                deep[i][j] = new StringBuilder(original[i][j].toString());
            }
```

```
        }

        // Modifying the deep copy
        deep[0][0].append(" Full Stack");

        // Printing the Original Array
        System.out.println("Original Array:");
        printArray(original);

        // Printing the Deep Copy
        System.out.println("\nDeep Copy:");
        printArray(deep);
    }

    // Method to print a 2D array
    static void printArray(StringBuilder[][] array)
    {
        for (int i = 0; i < array.length; i++)
        {
            for (int j = 0; j < array[i].length; j++)
            {
                System.out.print(array[i][j] + "\t");
            }
            System.out.println();
        }
    }
}
```